

# BROWSER TYCOON

## Product Requirements Document

*Chrome Extension | Idle Clicker Game*

Version 1.1 | May 2026

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# 1. Product Overview

## 1.1 Summary

Browser Tycoon is a Chrome extension that transforms everyday browsing into an idle clicker game. Inspired by Adventure Capitalist, it rewards real browsing behavior — opening tabs, visiting websites, staying on pages — with in-game currency (\$). Players assign domains to upgradeable slots, spend currency on per-domain upgrades, unlock more slots, and prestige for permanent bonuses.

Core design philosophy: the game rewards your existing browsing habits. You never need to browse differently. Over time, the sites you visit every day feel like investments.

## 1.2 Target Audience

- Casual gamers who browse frequently and enjoy incremental/idle game loops
- Power users and developers who spend significant time in Chrome
- Players of idle games (Adventure Capitalist, Cookie Clicker, etc.)

## 1.3 Platform & Distribution

| Item         | Detail                                                                                                            |
|--------------|-------------------------------------------------------------------------------------------------------------------|
| Platform     | Google Chrome (Manifest V3 extension)                                                                             |
| Distribution | Chrome Web Store (public listing)                                                                                 |
| Storage      | Two-tier: chrome.storage.local (full library) + chrome.storage.sync (hot data only — see Section 6)               |
| Background   | chrome.alarms (passive income while browser is closed)                                                            |
|              | <div><div>New Tab Integration</div><div>opt-in New Tab override via chrome_url_overrides in manifest.</div></div> |

## 1.4 Currency

The in-game currency is displayed as \$ (dollars). As values grow large, abbreviated suffixes are used — matching the Adventure Capitalist convention:

| Threshold       | Display Format |
|-----------------|----------------|
| < 1,000         | \$999          |
| 1,000 – 999,999 | \$1.23K        |
| 1M – 999M       | \$1.23M        |

|           |                                      |
|-----------|--------------------------------------|
| 1B – 999B | \$1.23B                              |
| 1T – 999T | \$1.23T                              |
| 1Qa+      | \$1.23Qa, Qi, Sx, Sp, Oc, No, Dc ... |

## 2. Core Mechanics

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### 2.1 How \$ Is Earned

Players earn \$ through two main streams:

#### Active Income (tab open)

- Every slotted domain generates \$/sec while its tab is open in the browser
- Rate is determined by the domain's base rate multiplied by applicable upgrades
- Focus Bonus applies when the tab is the currently active (foreground) tab
- Background Hum applies when the tab is open but not in focus
- Anti-Cheat: Only one tab per unique domain contributes to background income. If a domain is open in multiple non-active tabs, only one instance will accrue background income. If any tab for that domain becomes active, it counts as active income, and there is no concurrent background bonus from other tabs of the same domain.

#### Vault Payouts (on visit)

- Every slotted domain accumulates stored \$ while it is not being visited
- Vault accumulates even when the browser is closed (real-time clock delta)
- Vault has a cap determined by upgrade levels; stops accruing when full
- On the user's first visit to a domain each day, two bonuses are paid out:
  - Daily First-Open Bonus — flat bonus, scales with upgrade levels
  - Vault Payout — accumulated stored \$, separate from the daily bonus
- Both are displayed simultaneously in the popup: e.g. Daily: +\$1.2K • Vault: +\$8.4K

### 2.2 Base Income Rate

- Base rate is identical for all domains regardless of domain type
- All income multipliers are applied on top via upgrades
- A brand-new domain with no upgrades still earns a small trickle to onboard new players

 *Exact base rate value is defined in Section 8 (Growth & Balance System).*

### 2.3 Domain Slots

Slots are the central strategic resource. Only domains assigned to a slot generate income. The player starts with 3 slots and can unlock more.

#### Slot Behavior

- Player manually assigns any discovered domain to an open slot
- Domains can be swapped out of slots at any time with a short cooldown after removal. **The swap cooldown is one swap per slot per day.**
- Swapping a domain out clears its vault and the new domain swapped starts with no vault and no first day bonus.
- Removing a domain from a slot does not reset its upgrades — they persist in the Domain Library
- Each slot can independently be upgraded with prestige currency (see Section 4)

## Slot Unlock Costs (\$ — tuned per Section 8)

| Slot # | Unlock Cost | Notes                                                   |
|--------|-------------|---------------------------------------------------------|
| 1 – 3  | Free        | Starting slots                                          |
| 4      | \$500       | First unlock                                            |
| 5      | \$2,500     |                                                         |
| 6      | \$10,000    |                                                         |
| 7      | \$50,000    |                                                         |
| 8+     | Exponential | $\text{cost}(n) = 500 \times 5^{(n-4)}$ — see Section 8 |

**i** Slot unlock costs use \$ (not prestige currency) for new slots. **Slots for which Tier I (or higher) has been purchased with CP are permanently unlocked and persist through prestige resets.** Slots that have never been tiered must be re-unlocked with \$.

## 2.4 Domain Library

The Domain Library is a persistent store of every domain the player has ever visited or slotted. It acts as the player's portfolio and uses a two-tier storage model to stay within Chrome's sync limits.

### Two-Tier Library Model

| Tier             | Triggered By                               | Data Stored                                   |
|------------------|--------------------------------------------|-----------------------------------------------|
| Seen Record      | Any domain visit                           | domain name + firstSeen timestamp (~50 bytes) |
| Full DomainEntry | Domain is slotted OR any upgrade purchased | All fields in schema below (~250 bytes)       |

**i** A domain is promoted from Seen to Full automatically the moment the player slots it or buys its first upgrade. Demotion never occurs — once promoted, always Full.

### Full DomainEntry Fields

- domain — string (e.g. "youtube.com")
- upgrades — Record<upgradeId, level> — upgrade levels per category
- vaultAmount — number — current stored \$
- vaultLastTickTime — number — when vault last accrued
- lastVisited — number — Unix timestamp
- lifetimeEarned — number
- dailyBonusClaimedDate — ISO date string of last daily claim
- isSlotted — boolean
- slotId — number | null

### Library Behavior

- Vault does NOT accumulate for unslotted domains — only slotted domains accrue vault income

- If a domain with an existing vault balance is re-slotted and then visited, the vault pays out normally
- Library is searchable and filterable (by name, upgrade level, last visited, tier)

## 3. Domain Upgrades

Every domain shares the same upgrade tree. Upgrade levels are fully independent per domain — upgrading YouTube's Tab Multiplier does not affect GitHub's. Upgrades are purchased with \$. Costs scale exponentially per level (see Section 8 for exact curves).

### 3.1 Category A — Active Income

| Upgrade        | Effect                                                      | Max Level      | Growth Rate |
|----------------|-------------------------------------------------------------|----------------|-------------|
| Tab Multiplier | +X% \$/sec while tab is open                                | N/A (Infinite) | 1.6×        |
| Focus Bonus    | Extra multiplier when this tab is the active foreground tab | N/A (Infinite) | 1.7×        |
|                |                                                             |                |             |
|                |                                                             |                |             |

### 3.2 Category B — Vault & Passive Storage

| Upgrade           | Effect                                                                                                       | Max Level      | Growth Rate |
|-------------------|--------------------------------------------------------------------------------------------------------------|----------------|-------------|
| Cold Storage      | Increases the cap on how much \$ the vault can hold while unvisited                                          | N/A (Infinite) | 1.6×        |
| Storage Duration  | Extends how long the vault keeps accumulating before it is full                                              | N/A (Infinite) | 1.7×        |
| Compound Interest | Stored \$ grows a small % over time while sitting in the vault (0.2% per minute per level, capped at 1%/min) | 5              | 2.0×        |
| Windfall Bonus    | One-time burst added on top of vault payout; scales with time since last visit                               | N/A (Infinite) | 2.0×        |

### 3.3 Category C — Background Behavior

| Upgrade        | Effect                                                                          | Max Level      | Growth Rate |
|----------------|---------------------------------------------------------------------------------|----------------|-------------|
| Background Hum | Domain earns a % of its active income even when the tab is open but not focused | N/A (Infinite) | 1.6×        |
| Idle Depth     | Background income increases the longer the tab sits idle                        | N/A (Infinite) | 1.8×        |
| Wake Bonus     | \$ burst when the user switches back to a backgrounded tab                      | N/A (Infinite) | 2.0×        |

### 3.4 Income Formula

The following formula defines per-domain \$/sec:

```
income_per_sec =
  BASE_RATE
  × tab_multiplier_bonus(level)           // upgrade curve, Category A
  × focus_bonus(level)                   // only if active foreground tab
    OR × background_hum_pct(level)       // only if tab open but not focused
  × slot_tier_bonus(slot.tier)           // applied last, multiplicative
```

Vault payout on first visit:

```
payout =
  min(vault_amount, cold_storage_cap(level))
  × compound_interest_factor(minutes_idle, level)
  + windfall_bonus(hours_since_visit, level)
  + daily_first_open_bonus(upgrade_levels) // always applies on first daily visit
```

### 3.5 Global Upgrades

These upgrades apply globally to the player's browsing activity. They are split into two categories based on currency: \$ upgrades are tactical and reset on prestige; CP upgrades are permanent, character-level investments that persist through prestige.

| Upgrade                         | Currency | Reasoning                                            |
|---------------------------------|----------|------------------------------------------------------|
| Tab count multiplier (< 3 tabs) | CP       | Permanent commitment to focused browsing.            |
| Tab count multiplier (3–7 tabs) | CP       | Mid-range baseline playstyle.                        |
| Tab count multiplier (8+ tabs)  | CP       | Permanent commitment to power-user identity.         |
| Browser search bonus            | \$       | Tactical; cheap, early-run global spending category. |
| Global \$/sec multiplier (flat) | CP       | Permanent backbone income upgrade.                   |
| Bookmark visit bonus            | \$       | Run-specific flavor, easy to rebalance.              |
| Incognito penalty/immunity      | CP       | Permanent quality-of-life unlock.                    |

## 4. Prestige System

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### 4.1 Overview

Prestige ("Clear Cache") allows players to reset their progress in exchange for Cache Points (CP), the prestige currency. CP is used to permanently upgrade individual slots and unlock later slot counts.

### 4.2 What Resets on Prestige

- All \$ balance
- All domain upgrade levels (across all domains in the library)
- Vault amounts reset to zero
- Session/streak progress

### 4.3 What Persists Through Prestige

- Number of unlocked slots
- Slot tier upgrade levels (bought with CP)
- Domain Library entries (domain names, tier badges, lifetime earnings history)
- Full DomainEntry promotions — once promoted, domains stay promoted
- Total Cache Points previously spent (slot tiers are already applied)

### 4.4 Cache Points Formula

CP awarded on prestige is based on lifetime \$ earned (totalEarned) at the time of reset. Using a square-root curve (matching the AdVenture Capitalist model) gives diminishing returns that reward going further each run without making early resets worthless.

```
CP_gained = floor( sqrt( totalEarned / PRESTIGE_DIVISOR ) )
```

```
PRESTIGE_DIVISOR = 1,000,000    (tune during balance pass)
```

```
Example: totalEarned = $25M → floor(sqrt(25)) = 5 CP
```

```
Example: totalEarned = $100M → floor(sqrt(100)) = 10 CP    (2× CP requires 4× earnings)
```

Design properties of this curve:

- To double CP, the player must earn ~4× as much as the previous run — rewards going further, not spam-resetting
- Lifetime earnings (not since-last-reset) means slow runs still contribute — appropriate for a passive browsing game
- PRESTIGE\_DIVISOR is the primary balance lever: lower = more frequent rewarding resets; higher = longer runs per cycle

 *Open design question: per-domain prestige vs. global reset only. Currently scoped as global reset only.*

### 4.5 Slot Tier Upgrades (Prestige Currency)

Each slot can be individually upgraded using CP. The upgrade belongs to the slot, not the domain — swapping a domain into an upgraded slot immediately benefits from the tier bonus.

| Tier               | CP Cost | Income Bonus                               | Notes                                                                     |
|--------------------|---------|--------------------------------------------|---------------------------------------------------------------------------|
| I                  | 1 CP    | +10% to all income from this slot          | First tier purchase permanently unlocks this slot across prestige resets. |
| II                 | 3 CP    | +25%                                       |                                                                           |
| III                | 8 CP    | +50%                                       |                                                                           |
| IV                 | 20 CP   | +100% (2×)                                 | Major power spike                                                         |
| V                  | 50 CP   | +200% (3×)                                 | End-game slot                                                             |
| Daily Streak Bonus | CP      | Increase Daily First-Open Bonus multiplier | This upgrade is tied to the slot and persists through prestige.           |

 Slot tier bonuses are multiplicative and applied after all domain upgrade calculations.

## 5. Player Progression & Strategies

### 5.1 Behavioral Progression Phases

Unlike traditional idle games where stages are gated by \$ thresholds, Browser Tycoon stages are driven by browsing behavior. The phases below are organic, not enforced — their duration varies by how much and how diversely the player browses.

| Phase          | Typical Timing              | What's Happening                                                                                                                         |
|----------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1 — Discovery  | Days 1–3                    | Player slots new domains frequently, hits first-visit bonuses, learns the loop. Base income feels small — novelty is the hook.           |
| 2 — Portfolio  | Week 1–2                    | Player identifies their core 5–8 domains and concentrates upgrades. Decisions about which domains to invest in become meaningful.        |
| 3 — Vault Loop | Week 2+                     | Active income well-upgraded. Player starts caring about overnight vault payouts and morning collection rituals.                          |
| 4 — Prestige   | When upgrade trees near max | Several upgrades maxed, slot unlocks expensive. Natural push toward Clear Cache. Run 2 starts with library intact and slot tiers active. |

**i** A player who browses 40+ sites/day will reach Phase 2 much faster than one who uses only 4–5 domains. The progression floor in Section 5.3 addresses this variance.

### 5.2 Progression Loop

- Browse normally → domains discovered and added to Library as Seen Records
- Assign top domains to slots → begin earning \$/sec → domain promoted to Full DomainEntry
- Spend \$ on domain upgrades → increase income
- Unlock additional slots → more domains earning simultaneously
- Return to browser → collect vault payouts and daily bonuses
- Prestige → earn CP, reset \$, upgrade slots permanently
- Repeat with stronger slot tiers → faster income scaling per run

### 5.3 Progression Floor (Low-Diversity Browsing)

Players who visit only a few domains must still have a meaningful game. The following safeguards prevent stagnation:

- Slot unlock costs gate on library size — slot 4 only offers to unlock once the library has at least 4 promoted domains
- Subdomains count separately (app.github.com vs github.com) to create natural discovery within focused browsing
- The Vault system ensures offline/light-browsing sessions still feel productive via morning collection rituals
- A weekly milestone bonus rewards players regardless of unique domain count (flat \$ bonus, resets Monday)

## 5.4 Emergent Player Strategies

| Strategy          | Description                                                                                                                          |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| The Grinder       | All slots filled with daily-use sites. Maximize Tab Multiplier and Focus Bonus upgrades. Earns most \$ from active focused browsing. |
| The Banker        | Mix active sites with rarely-visited domains that have high vault upgrades. Check in periodically to collect large vault payouts.    |
| The Night Owl     | Slots filled with backgrounded sites. Max Background Hum and Idle Depth. Earns passively while working in other tabs.                |
| The Collector     | Rapidly discovers new domains for the one-time discovery burst. Prioritizes breadth over depth.                                      |
| The Slot Investor | Pushes specific slots to Tier IV–V with CP, then carefully places highest-earning domains in those slots for maximum returns.        |

## 6. Technical Architecture

### 6.1 Chrome APIs Used

| API                        | Purpose                                                                                                 |
|----------------------------|---------------------------------------------------------------------------------------------------------|
| chrome.tabs                | Detect tab open/close/activate events in real time                                                      |
| chrome.webNavigation       | Track domain visits and page navigation within a domain                                                 |
| chrome.storage.local       | Full Domain Library — 10MB limit, not synced across devices                                             |
| chrome.storage.sync        | Hot game state only (balance, CP, slots, promoted domain upgrades) — 100KB limit, synced across devices |
| chrome.alarms              | Fire background ticks every minute for vault accrual while popup is closed                              |
| chrome.action.setBadgeText | Display current \$/sec or vault-ready indicator on the extension icon                                   |

### 6.2 Storage Architecture

Storage is split into two layers to stay within Chrome's sync limits while retaining the full domain history locally.

| Layer        | Store         | Limit  | Contents                                                                          |
|--------------|---------------|--------|-----------------------------------------------------------------------------------|
| Full Library | storage.local | 10 MB  | All Seen Records + all Full DomainEntries                                         |
| Hot Sync     | storage.sync  | 100 KB | balance, CP, unlocked slots, slot tiers, upgrade levels for promoted domains only |

**i** Vault amounts and tick timestamps are stored in local only. If a player switches devices, vault progress since the last sync is lost — this is an acceptable tradeoff. Strategy: sync vault amounts on a best-effort basis whenever the popup is opened.

### 6.3 Storage Schema

#### syncState object (chrome.storage.sync)

- balance — number — current \$ balance
- totalEarned — number — lifetime \$ earned (used for prestige calculation)
- cachePoints — number — prestige currency balance
- unlockedSlots — number — total slots available
- prestigeCount — number
- slots — Slot[] — array of slot objects
- promotedDomains — Record<domain, CompactUpgrades> — upgrade levels only, no timestamps

#### Slot object

- id — number
- tier — 0–5 — slot tier upgrade level
- assignedDomain — string | null
- swapCooldownUntil — number | null — Unix timestamp

### localState object (chrome.storage.local)

- seenDomains — Record<domain, { firstSeen: number }> — lightweight seen records
- domainLibrary — Record<domain, DomainEntry> — full entries for promoted domains only
- lastTickTime — number — Unix timestamp of last background tick

## 6.4 Background Tick Logic

A chrome.alarms alarm fires every minute. The service worker:

- Reads localState from storage
- For each slotted domain: calculates vault accrual since lastTickTime (capped at vault max)
- Applies Compound Interest growth to existing vault amounts
- Updates vaultAmount and vaultLastTickTime for each domain
- Writes updated localState back to storage

 *Active income (\$/sec while tab is open) is calculated in the popup's live JS loop, not in the background worker, to keep alarm load minimal.*

## 6.5 Cross-Device Sync Considerations

- chrome.storage.sync carries only upgrade levels and balances — no vault timestamps
- On a new device, vault amounts are initialized to zero; domains re-accrue from that point
- Promoted domain upgrades sync fully — a player's investment history follows them across devices
- If sync data approaches 80KB, prune CompactUpgrades entries for domains not visited in 90+ days

## 7. UI Overview (Mechanics-Focused)

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Exact visual design is deferred. The following screens and states must be supported by the UI:

### 7.1 Main Popup (380×580px)

- Current \$ balance (large, prominent) with abbreviated suffix
- \$/sec display
- Slot grid — shows all unlocked slots, each displaying assigned domain (favicon + name) or empty state
- Vault-ready indicator on slots where vault is full or daily bonus is unclaimed
- Quick access to Domain Library
- Prestige button (with confirmation dialog showing CP to be earned)

### 7.2 Slot Detail View

- Domain name and favicon
- Current income breakdown (active, focus, background, vault)
- Three upgrade category panels with per-upgrade current level, next level effect, and cost
- Swap domain button (with cooldown timer if active)

### 7.3 Domain Library View

- Seen Records shown as a browseable list (greyed, no upgrade UI)
- Full DomainEntry domains shown with tier badge, upgrade levels, and lifetime earned
- Filter/search bar — by name, tier, last visited, upgrade level
- Clear visual distinction between Seen and Promoted domains

## 8. Growth & Balance System

This section defines the mathematical framework governing all cost curves, income scaling, and the prestige loop. Values marked as [TUNE] are starting points to be adjusted during a dedicated balance pass — the formulas themselves are fixed design decisions.

### 8.1 Core Design Principle

Browser Tycoon uses the standard idle-game seesaw: upgrade costs grow exponentially while income grows polynomially. Most \$ upgrades scale infinitely to allow trillions of currency to be spent via the "Buy Max" feature. The tension between them creates the satisfying "just one more upgrade" loop. The key difference from games like AdVenture Capitalist is that domains are discovered, not purchased — so the player count of active generators (domains in slots) grows slowly and behaviorally, not as a direct purchase decision.

### 8.2 Upgrade Cost Curves

All upgrade costs follow a geometric series. Each upgrade category has its own base cost and growth rate, tuned so that different categories feel most valuable at different \$ thresholds (early, mid, and late run).

#### 8.2.1 Infinite Scaling & Buy Max

Most upgrades have no maximum level. The 'Buy Max' button allows players to instantly purchase levels until they hit the exponential cost wall. 'Compound Interest' is the notable exception, capped at Level 5.

```
upgrade_cost(level) = base_cost * growth_rate ^ level
```

Buying from level 0 → level N costs:

```
total_cost = base_cost * (growth_rate^N - 1) / (growth_rate - 1)
```

| Upgrade           | Base Cost    | Growth Rate | Peak Value | Design Intent                                  |
|-------------------|--------------|-------------|------------|------------------------------------------------|
| Tab Multiplier    | \$25 [TUNE]  | 1.6×        | Early run  | First upgrade most players buy; cheap early    |
| Focus Bonus       | \$40 [TUNE]  | 1.7×        | Early-mid  | Rewards active tab use                         |
|                   |              |             |            |                                                |
|                   |              |             |            |                                                |
| Cold Storage      | \$60 [TUNE]  | 1.6×        | Mid run    | Becomes valuable once vault rhythm established |
| Storage Duration  | \$100 [TUNE] | 1.7×        | Mid-late   | Pairs with Cold Storage                        |
| Compound Interest | \$200 [TUNE] | 2.0×        | Late run   | Luxury upgrade; expensive but passive          |
| Windfall Bonus    | \$250 [TUNE] | 2.0×        | Late run   | Rewards infrequent check-in strategy           |
| Background Hum    | \$50 [TUNE]  | 1.6×        | Early-mid  | Broad appeal; any browsing pattern benefits    |

|            |              |      |          |                                   |
|------------|--------------|------|----------|-----------------------------------|
| Idle Depth | \$90 [TUNE]  | 1.8× | Mid run  | Night Owl strategy cornerstone    |
| Wake Bonus | \$150 [TUNE] | 2.0× | Late run | Small luxury; tab-switcher reward |

### 8.3 Upgrade Effect Values

These define what each level of an upgrade actually does — the multiplier or bonus granted per level:

```
// Category A - multiplicative income bonuses
tab_multiplier_bonus(level) = 1 + (0.15 × level)    // +15% per level, e.g. L5 = 1.75×
focus_bonus(level)         = 1 + (0.20 × level)    // +20% per level

// Category B - vault bonuses
cold_storage_cap(level)     = BASE_RATE × 3600 × (1 + level) // hours of income
compound_interest_factor(min, lvl) = (1 + 0.002 × lvl) ^ min // 0.2%/min per level
windfall_bonus(hours, lvl)  = BASE_RATE × hours × 0.1 × lvl

// Category C - background bonuses
background_hum_pct(level)   = 0.05 × level    // 5% of active income per level
idle_depth_bonus(t, lvl)    = background_hum_pct × (1 + 0.1 × lvl × min(t/300, 5))
                                // ramps over 25 min of idle time
wake_bonus(lvl)             = BASE_RATE × 30 × lvl // 30 seconds of income per level
```

### 8.4 Base Rate

BASE\_RATE is the foundation all formulas build on. A new unupgraded domain in a standard slot should feel like a meaningful but small trickle.

```
BASE_RATE = $0.10 / sec    [TUNE]

// A fully maxed domain in a Tier V slot at max Tab Multiplier + Focus Bonus:
// 0.10 × (1 + 1.5) × (1 + 2.0) × 3.0 (slot) ≈ $2.33/sec
// Across 8 such slots ≈ $18.60/sec
// This should feel achievable only near prestige threshold
```

### 8.5 Slot Unlock Cost Curve

Slot unlocks use a steeper exponential to make each slot feel like a significant milestone:

```
slot_unlock_cost(n) = 500 × 5^(n - 4)    for n ≥ 4
```

```
Slot 4:  $500
Slot 5:  $2,500
Slot 6:  $12,500
Slot 7:  $62,500
Slot 8:  $312,500
...
```

## 8.6 Prestige Math

The CP formula uses a square root of lifetime earnings, matching the AdVenture Capitalist model. This ensures:

- Diminishing returns — each successive run must go further to earn the same CP gain
- Lifetime-based — slow days still contribute; appropriate for a passive browsing game
- Regular cycles encouraged — waiting indefinitely yields diminishing marginal CP per hour

```
CP_gained = floor( sqrt( totalEarned / PRESTIGE_DIVISOR ) )
```

```
PRESTIGE_DIVISOR = 1,000,000 [TUNE]
```

To double CP per run, the player needs ~4× the earnings of the prior run.

To earn 10 CP: totalEarned must reach \$100M.

To earn 20 CP: totalEarned must reach \$400M (4× more, for 2× the CP).

Run acceleration: because the Domain Library and slot tier bonuses persist, each successive run should feel noticeably faster in Phase 1 and 2. A Tier III slot (50% bonus) effectively compresses the time to reach the same \$ milestone by roughly 33%. The balance goal is that run 2 should reach the prior run's prestige threshold in roughly 60–70% of the time.

## 8.7 Vault Math

The vault is a capped passive accumulator. Its rate is a fraction of active income so it never trivializes the active loop:

```
vault_rate = BASE_RATE * 0.5    // vault accrues at 50% of base rate [TUNE]

vault_amount += vault_rate * elapsed_seconds
vault_amount  = min(vault_amount, cold_storage_cap(level))

// Compound Interest applied during background tick (once per minute):
vault_amount *= (1 + 0.002 * compound_interest_level)
vault_amount  = min(vault_amount, cold_storage_cap(level)) // re-cap after growth
```

**i** Compound Interest rate (0.2%/min at L1) is intentionally small to prevent vault from dominating income. At max level (L5, 1%/min), a full 8-hour overnight vault grows by ~394% — meaningful but not game-breaking.

### 8.8 Balance Pass Checklist

Before shipping, validate the following targets with a spreadsheet simulation:

| Target                                   | Expected Value                                    |
|------------------------------------------|---------------------------------------------------|
| Time to first slot unlock (Slot 4)       | ~15–20 min of active browsing [TUNE]              |
| Time to fill 6 slots                     | ~3–5 days of normal browsing [TUNE]               |
| Time to first prestige                   | ~7–14 days [TUNE]                                 |
| CP earned on first prestige              | 3–8 CP (enough for 2–3 slot tier upgrades) [TUNE] |
| Run 2 speed vs Run 1                     | ~60–70% of the time to reach same milestone       |
| Max \$/sec (fully upgraded, Tier V slot) | ~\$2–5/sec per domain [TUNE]                      |
| Vault overnight payout (8 hrs, maxed)    | Equivalent to ~2–4 hrs of active income [TUNE]    |

## Appendix A — Open Design Questions

| Question                       | Current Decision                                                                              |                                                                                             |
|--------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Per-domain vs. global prestige | Global reset only for v1.0. Per-domain prestige may be explored in a later version.           |                                                                                             |
| Subdomain counting             | Subdomains count separately (app.github.com ≠ github.com) to help low-diversity browsers.     |                                                                                             |
| Weekly milestone bonus         | Flat \$ bonus on Monday reset, amount TBD. Rewards players regardless of unique domain count. |                                                                                             |
| Sync conflict resolution       | Last-write-wins on sync. Vault amounts are local-only to avoid double-counting.               |                                                                                             |
| Library pruning                | Seen Records older than 180 days with no upgrades may be pruned from local storage.           |                                                                                             |
|                                | Daily bonus reset time                                                                        | <b>Daily resets (including the Daily First-Open Bonus) occur at midnight local time.</b>    |
|                                | Incognito tabs                                                                                | <b>Incognito tabs do not count towards income, domain discovery, or any game mechanics.</b> |
|                                |                                                                                               |                                                                                             |

## 9. Theme and Design

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### 9.1 Aesthetic Direction

The game will use a **Cyberpunk / Terminal Pixel Art Aesthetic** defined by a dark teal/navy background, cyan and purple accent colors, and a pixel font like “Press Start 2P” or similar. The design will leverage CSS for scan-line overlays, glow effects, and stylized borders.

### 9.2 Visual Assets & Strategy

Domains will retain their unique **favicons** for personalization, with a clear fallback for failure. Custom art will be reserved for game chrome, including: Upgrade Icons (sourced from open-source packs like Iconify or Kenney.nl, focusing on a tech aesthetic of circuit boards and coils); Slot Frames (CSS only); Vault-Ready Indicators (CSS animation); Tier Badges (CSS); and Prestige Animation (a full-screen CSS/JS glitch effect/screen wipe).

### 9.3 New Tab Page Integration

The extension will feature an **opt-in New Tab override that shows** a minimal game dashboard to provide a passive re-engagement moment. This lightweight dashboard will display: Current balance and \$/sec, a compact view of slots with vault-ready indicators, and the closest milestone progress bar. Implementation uses the `chrome_url_overrides` manifest setting and must be explicitly enabled by the user.

### 9.4 Onboarding Flow

To address high churn at install, an explicit onboarding flow will be implemented. This should be a guided walk-through after installation, explaining the core loop, slots, earning mechanisms, and upgrades.

— End of Document —